

4.3 Contrastive Learning and Zero-Shot Transfer

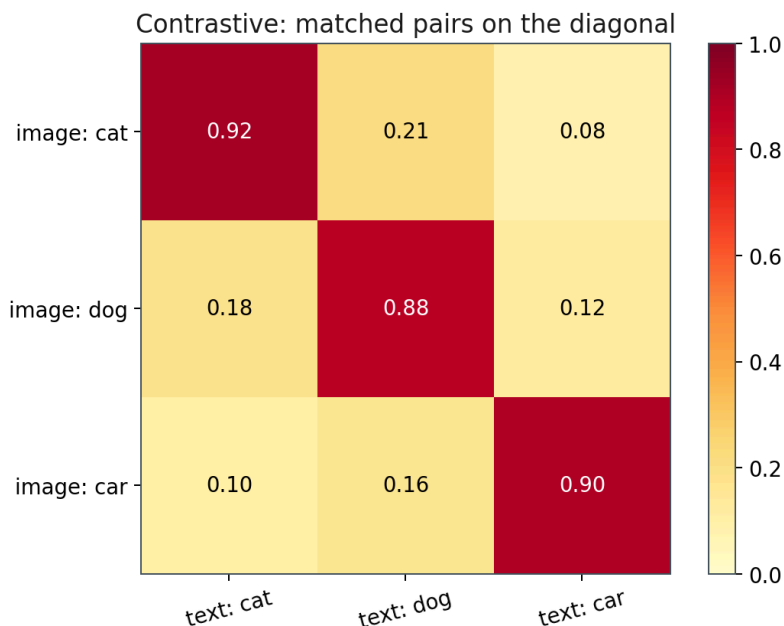
Matched pairs on the diagonal. Class names as text embeddings.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Contrastive learning pulls matched pairs together and pushes unmatched pairs apart, in a shared space. **Zero-shot** transfer is what you get when one side of the pair can be a class name written as text.

1. A batch of matches

N images and N captions. Encode both. Cosine similarities form an $N \times N$ matrix. The loss (InfoNCE) wants the **diagonal** hot: image i with caption i , not caption j .

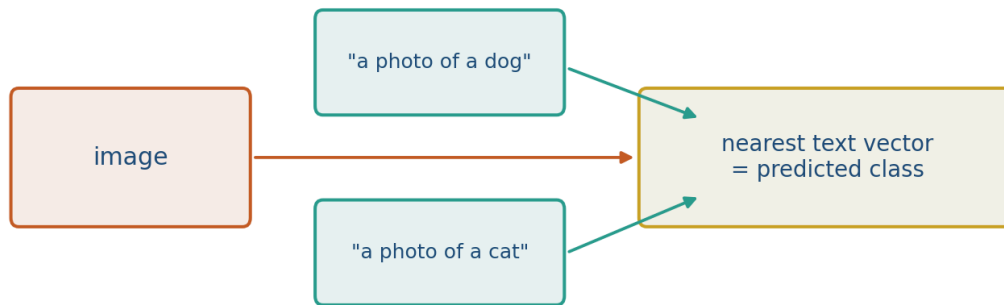


This is coordinated representation (note **4.1**) with a specific loss. You never fuse pixels and words into one vector during pretraining; you only score pairs.

2. Zero-shot classification

At test time, encode the image once. Encode prompts "a photo of a dog", "a photo of a cat", ... Take the nearest prompt. There is no trained softmax over ImageNet. New classes are new strings.

Zero-shot: class names are text embeddings, not trained heads



Prompt wording matters (a photo of vs the raw label). So does language: a class that the text tower never saw as a phrase will not magically appear.

3. What can go wrong

The negative set is the rest of the batch. Small batches $\Rightarrow$ easy negatives $\Rightarrow$ a weak space. Web pairs are noisy (Week 5: BLIP filters them). Contrastive spaces can still be **biased** (Week 5.3): "doctor" sits nearer to one gender's photos.

4. Practice

1. If every caption in the batch is the word *photo*, what happens to the diagonal?
2. Why is "a photo of a golden retriever" often better than "golden retriever" for CLIP-style zero-shot?
3. Name one class you **cannot** add at test time by typing it.